

Exact Standard-Channel Rank of a Signed-Reversal Incidence Layer

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Abstract

We study a noncentral signed-reversal incidence layer in the signed permutation group. For the diagonal layer $\tilde{B}I_n(k, k)$, the aggregate operators on the two ambient standard channels have exact ranks $\lceil n/2 \rceil$ and $\lceil n/2 \rceil - 1$ whenever the layer is nonempty. The rank numbers are classical matching-scheme multiplicities; the object-specific content is the collapse of the signed-reversal layer onto the reversal idempotent and the positivity of the resulting scalars. We also record a positive-only separator add-on: for $X_n = \tilde{B}I_n(1, 1)$ and $n \geq 4$, the layer is neither a signed-poset linear-extension set nor an exact left-translate tiler of B_n .

Keywords: signed permutations; hyperoctahedral group; reversal incidence layers; association schemes; rook polynomials; signed posets.

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1 Introduction

This paper studies a signed analogue of the type-A one-incidence layer. In type A, the layer is defined by one hit on the identity diagonal and one hit on the reflected diagonal determined by the reversal permutation. Here the ambient group is the signed permutation group B_n , and the signed layer records positive hits on the identity diagonal and on the reversal diagonal. The reversal diagonal is essential: it is the geometric second diagonal that makes the object noncentral but still rigid enough to be controlled by the matching scheme of the reversal pairs.

The main theorem is deliberately scoped. We do not attempt a full native B_n Fourier fingerprint, a Type-D analogue, or a general Weyl-group theorem. We prove the exact ranks of two ambient standard-channel aggregate operators on every nonempty diagonal row $\tilde{B}I_n(k, k)$. The numbers that appear, $\lceil n/2 \rceil$ and $\lceil n/2 \rceil - 1$, are classical multiplicities from the matching scheme attached to the reversal involution. The object-specific contribution is that this noncentral signed-reversal layer collapses onto the corresponding matching-scheme components, and that the remaining scalars are strictly positive.

There are two independent additions around the main theorem. First, the positive-only row $X_n = \tilde{B}I_n(1, 1)$ is separated from two natural chamber families: for $n \geq 4$ it is neither a Reiner-style signed-poset linear-extension set nor an exact left-translate tiler of B_n . Second, the appendix records finite $n = 8$ native B_n fingerprint evidence. That evidence explains the known zero rows in one finite atlas slice, but it is not promoted to a classification theorem.

The proof of the rank theorem has three parts. The even rows collapse directly by centralizer symmetry and a permanent-domination positivity lemma. The odd rows reduce to two residual scalar sequences, called H_m and K_m . These residuals are identified with signed-rook moment formulae and then proved positive by exact order-three recurrences and ratio induction. The computational material in the repository verifies the polynomial identities and finite edge rows using exact integer or rational arithmetic; the paper states the mathematical identities that those certificates check.

Structure. Section 2 fixes the signed-reversal layer and states the main theorem. Section 3 proves the positive-only separator theorem. Sections 4–9 prove the rank theorem. Section 10 records the claim boundary. The appendices give the exact replay routing and the finite native B_n fingerprint boundary.

2 Definitions and Main Statement

Let B_n be the signed permutation group, written as pairs (π, ε) with $\pi \in S_n$ and $\varepsilon_i \in \{\pm 1\}$. We use zero-indexed coordinates and set

$$\rho(i) = n - 1 - i.$$

Define

$$\tilde{B}I_n(k, \ell) = \{(\pi, \varepsilon) \in B_n : \#\{i : \pi(i) = i, \varepsilon_i = +1\} = k, \#\{i : \pi(i) = \rho(i), \varepsilon_i = +1\} = \ell\}.$$

The theorem concerns only the diagonal layer $\tilde{B}I_n(k, k)$ and only the two ambient standard-channel aggregate operators M_{ref} and $M_{\text{pair-std}}$.

Theorem 1 (Main theorem). *For every $n \geq 2$ and every $k \geq 1$ such that $\tilde{B}I_n(k, k)$ is nonempty,*

$$\text{rank } M_{\text{ref}}(\tilde{B}I_n(k, k)) = \left\lceil \frac{n}{2} \right\rceil, \quad \text{rank } M_{\text{pair-std}}(\tilde{B}I_n(k, k)) = \left\lceil \frac{n}{2} \right\rceil - 1.$$

Remark 1 (Boundary). *No Type-D theorem, full B_n fingerprint theorem, enumeration novelty claim, or general Weyl theorem is asserted. The positive-fixed marginal is treated as classical signed-permutation enumeration; in particular the positive-fixed zero row is the type-B derangement sequence [OEI26, CZ07]. The rank number and the reversal projector are matching-scheme bookkeeping [Del73, BI84, GR01]. The contribution here is the collapse and scalar positivity for the specific noncentral layer $\tilde{B}I_n(k, k)$, plus the separate separator theorem below for $\tilde{B}I_n(1, 1)$.*

3 Positive-Only Separator Add-On

Set

$$X_n = \tilde{B}I_n(1, 1).$$

This section uses the equivalent one-indexed signed-word convention $w = (w_1, \dots, w_n)$, with $|w_1|, \dots, |w_n|$ a permutation of $\{1, \dots, n\}$ and $\rho(i) = n + 1 - i$. Thus

$$X_n = \{w \in B_n : \#\{i : w_i = i\} = 1, \#\{i : w_i = \rho(i)\} = 1\}.$$

For a signed word define the Reiner-style total order [Rei93]

$$T(w) : -w_n < \dots < -w_1 < 0 < w_1 < \dots < w_n.$$

Let $\Sigma_n = \{-n, \dots, -1, 0, 1, \dots, n\}$. For $X \subseteq B_n$, put

$$C(X) = \bigcap_{w \in X} \{(a, b) \in \Sigma_n^2 : a \text{ precedes } b \text{ in } T(w)\}.$$

If $X = L_B(P)$ is the signed linear-extension set of a signed poset P , then every strict relation of P , and every relation in its transitive closure, lies in $C(X)$.

Theorem 2 (Positive-only separator). *For every $n \geq 4$, the layer $X_n = \tilde{B}I_n(1, 1)$ is not a Reiner-style signed-poset linear-extension set and is not an exact left-translate tiler of B_n . For $n = 3$, the divisibility obstruction disappears and there is an exact left-translate tiling of B_3 by X_3 .*

Proof. We first prove the signed-poset obstruction. For $n \geq 5$, any prescribed signs on at most two absolute labels can be extended to an element of X_n . Indeed, positive labels create fixed or reversal hits only when placed in their two dangerous rows, while negative labels create no positive hit. Place the prescribed positive labels, then add exactly one positive fixed hit and one positive reversal hit; all remaining labels are placed negatively. If the odd center is used, it is used as the intended center hit, since the center supplies both hit types at once. Otherwise auxiliary positive labels are chosen away from the center and away from their dangerous rows. With at most two prescribed labels, there is always such a spare choice for $n \geq 5$.

Given any ordered pair $a \neq b$ in Σ_n , choose signs so that b lies before 0 and a lies after 0 in $T(w)$. If a, b are nonzero and $a \neq -b$, prescribe $|a|$ with the sign of a and $|b|$ with the opposite sign to b . If $a = -b$, prescribe only $|a|$ with the sign of a . If one entry is 0, prescribe the other label on the required side of 0. The sign-extension construction then gives $w \in X_n$ in which b precedes a . Hence $C(X_n) = \emptyset$ for $n \geq 5$.

The boundary case $n = 4$ is finite: an exact finite check records witnesses in X_4 reversing all 72 ordered pairs in Σ_4 . Thus $C(X_4) = \emptyset$ as well. If $X_n = L_B(P)$ for some signed poset P , all relations of P would lie in $C(X_n)$, so P would be the antichain and $L_B(P) = B_n$. But $X_n \neq B_n$, for instance the all-negative identity has no positive fixed or reversal hit. This proves the signed-poset obstruction for all $n \geq 4$.

For the tiler obstruction, an exact left-translate tiling by X_n requires $N_n = |X_n|$ to divide $G_n = |B_n| = 2^n n!$. The count of X_n is obtained from the rook polynomial

$$R_{2m}(A, B, Z) = (1 + 2AZ + 2BZ + A^2Z^2 + B^2Z^2)^m,$$

$$R_{2m+1}(A, B, Z) = R_{2m}(A, B, Z)(1 + AZ + BZ + ABZ),$$

where A marks selected positive fixed events, B selected positive reversal events, and Z distinct forced cells. Inclusion-exclusion for exactly one event of each type gives

$$N_n = \sum_{p,q \geq 1, u} (-1)^{p+q} pq [A^p B^q Z^u] R_n(A, B, Z) 2^{n-u} (n-u)!.$$

Equivalently, for

$$S_n(Z) = (A\partial_A)(B\partial_B)R_n(A, B, Z)|_{A=B=-1},$$

we have

$$\frac{N_n}{G_n} = \sum_u \frac{[Z^u]S_n(Z)}{2^u (n)_u}.$$

The differentiated series are

$$S_{2m}(Z) = 4m(m-1)Z^2(1-Z)^2(1-4Z+2Z^2)^{m-2},$$

$$S_{2m+1}(Z) = Z(1-4Z+2Z^2)^m + 4m(m-1)Z^2(1-Z)^3(1-4Z+2Z^2)^{m-2}.$$

Substituting $Z = -x$ makes the tails alternating. The certificate `P15_S12_GATE_B_TILER_CERTIFICATE`, routed in Appendix A, records the exact finite window $4 \leq n \leq 21$, the $n = 3$ exact cover, and the symbolic all- n tail inequalities below.

For even $n = 2m \geq 22$, write

$$S_{2m}(-x) = 4m(m-1)x^2(1+x)^2f(x)^{m-2}, \quad f(x) = 1 + 4x + 2x^2.$$

Because $f(x) = (1 + (2 + \sqrt{2})x)(1 + (2 - \sqrt{2})x)$ has positive linear factors, the coefficients of $(1+x)^2f(x)^{m-2}$ are ultra-log-concave by Newton's inequalities. If $c_u = [x^u]S_{2m}(-x)$, this gives

$$\frac{c_{u+1}}{2^{u+1}(2m)_{u+1}} \leq \frac{c_u}{2^u(2m)_u},$$

so the normalized alternating tail is decreasing. The certificate then checks the two finite symbolic inequalities $L_9 > 1/11$ and $U_6 < 1/10$ after the shift $m = t + 11$; all coefficients of the resulting polynomials are positive.

For odd $n = 2m + 1 \geq 23$, the first coefficient of $S_{2m+1}(-x)$ is negative and all later coefficients are positive. Indeed

$$S_{2m+1}(-x) = xf(x)^{m-2}Q_m(x),$$

and the certificate lists the coefficients of Q_m : the nonconstant coefficients are positive for $m \geq 11$, while the top three support degrees are supplied by the positive high coefficients of Q_m . To prove the normalized tail is monotone from degree 2, put $C_m(x) = S_{2m+1}(-x)$ and use the equivalent coefficient positivity of

$$H_m(x) = 2((2m+1)C_m(x) - xC'_m(x)) - C_m(x)/x.$$

The identity $H_m(x) = f(x)^{m-3}P_m(x)$ is checked symbolically, and Newton log-concavity for $f(x)^{m-3}$ reduces the possible negative contribution of the linear term of P_m to a positive quadratic lower bound for every $m \geq 11$. As in the even case, the certificate checks $L_9 > 1/11$ and $U_6 < 1/10$ after the shift $m = t + 11$ by coefficient positivity. Thus the alternating-tail bounds and exact finite checks give

$$9 < G_4/N_4 < 10,$$

$$10 < G_n/N_n < 11 \quad (n \geq 5 \text{ odd}),$$

$$11 < G_n/N_n < 12 \quad (6 \leq n \leq 20 \text{ even}),$$

$$10 < G_n/N_n < 11 \quad (n \geq 22 \text{ even}).$$

Thus G_n/N_n is never an integer for $n \geq 4$, so X_n cannot tile B_n by exact left translates. For $n = 3$, an exact finite check records an exact cover by 24 left translates of the two-element set X_3 . $\square$

4 Proof Strategy and Matching-Scheme Background

Let P_ρ be the permutation matrix of the reversal involution and put

$$P_+ = \frac{I + P_\rho}{2}.$$

The centralizer of ρ in S_n acts on ordered coordinate pairs. For even $n = 2m$, its orbital algebra on coordinates is the matching, or cocktail-party, association scheme with basis $\{I, P_\rho, J - I - P_\rho\}$. For odd $n = 2m + 1$, the center coordinate adds the usual center row and column orbitals, but the non-center pair block is the same matching-scheme block. We use only the standard consequences that P_+ projects onto the $+1$ eigenspace of P_ρ , that

$$\text{rank } P_+ = \left\lceil \frac{n}{2} \right\rceil,$$

and that the corresponding pair-standard quotient has dimension $\lceil n/2 \rceil - 1$. These are classical association-scheme facts; the proof below supplies the layer-specific collapse onto these spaces and proves that the relevant scalars are nonzero.

Lemma 1 (Even collapse and positivity). *For even $n = 2m$ and every nonempty diagonal row $\tilde{B}I_{2m}(k, k)$ with $k \geq 1$, the reference-channel aggregate collapses to a positive scalar multiple of P_+ , and the pair-standard aggregate collapses to a positive scalar on the corresponding matching-scheme standard quotient. Hence the two even-row ranks are m and $m - 1$.*

Proof. The row is invariant under the centralizer of ρ , so the coordinate aggregates lie in the matching-scheme orbital algebra. In the reference channel, every generic off-pair orbit has a sign-reversing involution, while the diagonal and reversal orbits are equal by reversal symmetry. Thus only the P_+ component remains. The same block cancellation on ordered pair coordinates leaves only the pair-standard quotient component. The permanent-domination argument gives strict positivity of these surviving scalars on every nonempty even diagonal row, so the ranks are the classical multiplicities of the two matching-scheme components. $\square$

Lemma 2 (Odd structural reduction). *Let $n = 2m + 1$. The odd reference channel splits as*

$$M_{\text{ref}} = 2aP_+^{\text{pairs}} + a_c E_{\text{center}}.$$

The center split and trace identities reduce all nonempty odd reference rows except $k = 1$ to the even positive scalars and the center scalar. In the odd pair-standard channel, the bridge $\mu = a_c + 4a_{\text{even}}$ and the trace identity reduce all nonempty rows with $k \geq 2$ to the same positive inputs. Thus the only odd residuals are the reference scalar $a(2m + 1, 1) - a(2m, 1)$ and the pair-standard scalar $\lambda'(2m + 1, 1)$.

Proof. Separate the fixed center coordinate from the m non-center reversal pairs. The center coordinate contributes the one-dimensional block E_{center} , while the non-center coordinates remain in the matching-scheme block spanned by the pair idempotent P_+^{pairs} . Taking traces on the diagonal row expresses the non-center reference scalar in terms of the even scalar and the center contribution, except in the $k = 1$ row where the center-nonfixed correction is not eliminated. On the pair-standard side, the non-center pair quotient has two scalar pieces; the displayed bridge identifies one of them with already positive reference/even data, and the trace identity solves for the other for $k \geq 2$. The row $k = 1$ leaves precisely λ' . $\square$

5 Nonempty Diagonal Rows

Lemma 3 (Nonempty positive diagonal condition). *For the theorem range $k \geq 1$, the diagonal layer $\tilde{B}I_n(k, k)$ is nonempty exactly as follows. If $n = 2m$, then*

$$1 \leq k < m, \quad \text{or} \quad k = m \text{ and } m \text{ is even.}$$

If $n = 2m + 1$, then

$$1 \leq k \leq m, \quad \text{or} \quad k = m + 1 \text{ and } m \text{ is even.}$$

The row $k = 0$ is nonempty for every $n \geq 2$, but it is outside the theorem statement.

Proof. Prescribe two positive-hit sets: F for fixed hits and R for reversal hits. Off the odd center, a valid partial prescription must assign at most one image to each domain and must have distinct prescribed images. Thus a non-center position cannot simultaneously be a positive fixed hit and a positive reversal hit.

For $n = 2m$, this gives the upper bound $k \leq m$, since the k fixed-hit domains and k reversal-hit domains are disjoint. For $k < m$, construct examples by reversal-pair blocks. If $k = 2a$, choose a reversal pairs to be fully fixed-positive and a disjoint reversal pairs to be fully reversal-positive. If $k = 2a + 1$, do the same for $2a$ hits and use two further reversal pairs: one supplies a single positive fixed hit and the other supplies a single positive reversal hit. This uses k pairs when k is even and $k + 1$ pairs when k is odd, hence it is possible for every $k < m$.

At the top row $k = m$, all non-center positions are prescribed. If I is the set of positions sent as fixed, the complement is sent as reversed. Injectivity forces $I = \rho(I)$. Hence I is a union of reversal pairs and has even cardinality, so the top row exists exactly when m is even.

For $n = 2m + 1$, the center is fixed and reversed simultaneously. Give the center positive sign to contribute one fixed hit and one reversal hit. If $k = 1$, use the empty non-center prescription; if $k > 1$, apply the even $2m$ construction to the non-center positions with $k - 1$ hits of each kind. This gives exactly $1 \leq k \leq m$, plus the top row $k = m + 1$ when m is even. $\square$

6 Odd Reference Channel: the H_m Residual

For a polynomial $\sum_r h_r t^r$, write

$$\Phi_N \left(\sum_r h_r t^r \right) = \sum_r h_r 2^{N-r} (N-r)!.$$

This is the signed-rook completion functional, a signed version of the classical rook-polynomial completion count [KR46]: after r nonattacking special cells are selected in an N by N remaining board, the unmatched rows and columns can be completed by $2^{N-r} (N-r)!$ signed bijections.

Lemma 4 (H_m is the odd V_{ref} , $k = 1$, residual). *For $m \geq 2$,*

$$a(2m+1, 1) = a(2m, 1) + H_m.$$

The edge value is $H_2 = 22$. For $m \geq 3$, put

$$A(t) = 1 - 4t + 2t^2$$

and

$$P_m(t) = (2m-2)(2t-1)(2mt^3 - (4m+2)t^2 + (2m+2)t - 1) - A(t)^2.$$

Then

$$H_m = \Phi_{2m-1} \left((1-t)A(t)^{m-3} P_m(t) \right).$$

Proof. The odd reference operator splits the center coordinate from the non-center reversal pairs:

$$M_{\text{ref}} = 2aP_+^{\text{pairs}} + a_c E_{\text{center}}.$$

In the $k = 1$ row, the trace identity isolates the non-center scalar as

$$(2m)a(2m+1, 1) = (2m)a(2m, 1) + C_m,$$

where C_m is the center-nonfixed correction. It remains to compute C_m .

Force a non-center source and insert the odd center into one functional edge. The center split gives one fixed target, one reversal target, and $2m-2$ ordinary targets. After the forced edge is removed, the remaining board has $2m-1$ rows and columns, so its signed-rook expansion is evaluated by Φ_{2m-1} . The free non-center reversal pairs contribute the factor $A(t)$, and the two local insertion alternatives give

$$H_m = (2m-2)L_m - B_m,$$

with

$$B_m = \Phi_{2m-1} \left((1-t)A(t)^{m-1} \right),$$

$$L_m = \Phi_{2m-1} \left((1-t)(2t-1)A(t)^{m-3}(2mt^3 - (4m+2)t^2 + (2m+2)t - 1) \right).$$

Combining these two expressions gives the stated polynomial $P_m(t)$ and $C_m = 2mH_m$.

The only nonlocal algebra in the derivation is the equality between the expanded edge-insertion count and the compact Φ_{2m-1} expression. This is not a finite extrapolation step. The certificate `P15_S11A_H_TRUE_SCALAR_EDGE_INSERTION_VERIFIER`, routed in Appendix A, checks the local signed-rook edge factors, the transfer from the expanded expression to the displayed univariate target, and the following all- m annihilation.

After the small edge rows $m = 2, 3, 4$ are checked directly, the symbolic residual has the form

$$E_z(\text{local} - (2m-2)R_m) = A(s)^{m-5}Q_m(s), \quad m \geq 5,$$

where $E_z(F) = F(1, s) - \partial_z F(1, s)$. Under the moment substitution $s = 1/(2x)$, applying integration by parts to

$$I_j = \int_0^\infty e^{-x} (2x^2 - 4x + 1)^{m-5} x^j dx$$

gives the recurrence

$$jI_{j-1} + (-4j - 4m + 15)I_j + (2j + 4m - 12)I_{j+1} - 2I_{j+2} = -\delta_{j0}.$$

The verifier rewrites the transformed residual with this recurrence until only the basis $I_0, I_1, 1$ remains, and all three coefficients are identically zero in $\mathbb{Z}[m]$. Hence the displayed formula is the true scalar residual for every $m \geq 2$. $\square$

7 Odd Pair-Standard Channel: the K_m Residual

Lemma 5 (K_m is the odd pair-standard, $k = 1$, residual). *For odd $n = 2m + 1$, set*

$$K_m = (m - 1)\lambda'(2m + 1, 1).$$

The edge values are $K_2 = 28$ and $K_3 = 14640$. For $m \geq 4$, define

$$B(t) = 2t^2 - 4t + 1$$

and

$$\begin{aligned} L_m(t) = & 2(2m - 3)t + 2(-4m^2 - 8m + 19)t^2 + 2(24m^2 + 18m - 68)t^3 \\ & + 2(-56m^2 - 24m + 122)t^4 + 2(56m^2 + 24m - 92)t^5 \\ & + 2(-12m^2 - 44m + 12)t^6 + 2(-16m^2 + 56m)t^7 + 2(8m^2 - 24m + 8)t^8. \end{aligned}$$

Then

$$\lambda'(2m + 1, 1) = \Phi_{2m} \left(B(t)^{m-4} L_m(t) \right),$$

and therefore

$$K_m = (m - 1)\Phi_{2m} \left(B(t)^{m-4} L_m(t) \right).$$

Proof. The odd pair-standard channel has two scalar pieces, μ and λ' . The identity $\mu = a_c + 4a_{\text{even}}$ and the trace relation close every nonempty odd row with $k \geq 2$. Thus only the $k = 1$ scalar λ' remains.

Choose a distinguished non-center reversal pair and the odd center. Let j be a generic source outside that pair and outside the center. On the non-center pair-standard line,

$$\lambda' = M_{\text{pair}}[i, i] + M_{\text{pair}}[\rho(i), i] - 2M_{\text{pair}}[j, i].$$

Put $X = u - 1$, $Y = v - 1$, and $Z = uv - 1$, and define

$$\begin{aligned} a &= 1 + 2t(X + Y) + t^2(X^2 + Y^2), & c &= 1 + tZ, \\ sx &= 1 + tX, & sy &= 1 + tY, & h &= 1 + t(X + Y), \\ \text{same} &= (1 + u)sx + (1 + v)sy, & \text{generic} &= 4h^2. \end{aligned}$$

For the two same-pair entries, the forced edge lies inside the distinguished reversal pair. The center contributes c , the unused mate contributes sx or sy , and the other $m - 1$ non-center reversal pairs contribute a ; the two forced signs have weights $1 + u$ and $1 + v$. This gives $ca^{m-1}\text{same}$.

For the generic entry, the forced ordinary edge touches two non-center reversal pairs. Those two pairs leave two hook factors, giving h^2 ; the forced ordinary sign contributes 2, and the coefficient -2 in λ' gives the subtracted factor $generic = 4h^2$. The center again contributes c , and the remaining $m - 2$ reversal pairs contribute a^{m-2} . Since these row and column sets are disjoint, the rook polynomial is the product of the block factors. Therefore, for every $m \geq 2$,

$$\lambda'(2m+1, 1) = [uv] \Phi_{2m} \left(c \left(a^{m-1} same - a^{m-2} generic \right) \right).$$

A forced entry leaves $2m$ rows and columns, so the completion functional is Φ_{2m} .

The case $m = 1$ has no pair-standard quotient. For $m = 2, 3$, the product formula has nonnegative exponents but the univariate $B(t)^{m-4} L_m(t)$ presentation is not used; these are the direct edge rows $K_2 = 28$ and $K_3 = 14640$. For $m \geq 4$, coefficient extraction over $\mathbb{Z}[m, t]$ reduces the bivariate product above to $B(t)^{m-4} L_m(t)$. The public certificate route is split into three exact checks listed in Appendix A: `P15_S11A_K_TRUE_SCALAR_BIVARIATE_VERIFIER` checks the local true-entry factors, `P15_S11A_K_GLOBAL_PRODUCT_PHI_CERTIFICATE` checks the product/ Φ accounting over the independent reversal pairs, and `P15_S11A_SCALAR_LINK_SYMBOLIC_VERIFIER` checks that the bivariate target reduces to the displayed univariate $L_m(t)$. All residual polynomial coefficient lists are zero, so this proves that the displayed K_m sequence is the true pair-standard scalar sequence. $\square$

8 Ratio-Induction Positivity

Lemma 6 (Order-three positivity). *Let U_m denote either H_m or K_m , with an exact order-three recurrence*

$$p_0(m)U_m + p_1(m)U_{m+1} + p_2(m)U_{m+2} + p_3(m)U_{m+3} = 0.$$

Set $\rho_m = U_{m+1}/U_m$. If the required base rows, adjacent base-ratio checks, and polynomial sign checks hold, then $\rho_m \geq 4m^2$ propagates and $U_m > 0$ for all rows in the theorem range.

Proof. The recurrence gives

$$\rho_{m+2} = \frac{-p_2(m) - p_1(m)/\rho_{m+1} - p_0(m)/(\rho_m \rho_{m+1})}{p_3(m)}.$$

For the H_m recurrence, the base rows $m = 2, 3, 4$ have $H_m > 0$ and $\rho_m \geq 4m^2$, and the polynomial signs are $p_0 > 0$, $p_1 > 0$, $-p_2 > 0$, $p_3 > 0$, and

$$D_H(m) = 16m^2(m+1)^2(-p_2) - 4m^2p_1 - p_0 - 64m^2(m+1)^2(m+2)^2p_3 > 0.$$

This is exactly the cleared-denominator inequality for $\rho_{m+2} \geq 4(m+2)^2$ under $\rho_m \geq 4m^2$ and $\rho_{m+1} \geq 4(m+1)^2$.

For the K_m recurrence, the base rows $m = 2, \dots, 8$ have $K_m > 0$ and $\rho_m \geq 4m^2$, and for $m \geq 7$ the polynomial signs are $p_0 > 0$, $p_1 \leq 0$, $-p_2 > 0$, $p_3 > 0$, and

$$D_K(m) = 16m^2(m+1)^2(-p_2) - p_0 - 64m^2(m+1)^2(m+2)^2p_3 > 0.$$

The term $p_1 \leq 0$ can be discarded because it improves the lower bound. The adjacent base ratios ρ_7 and ρ_8 seed the recurrence from $m = 7$. Hence the same induction proves $K_m > 0$ for every $m \geq 2$. The recurrence identities are holonomic-style exact identities in the standard sense of creative telescoping and P-recursive verification [Zei90, PWZ96]; no finite extrapolation is used. In the replay table, `P15_S8_HM_RECURRENCE_CERTIFICATE` supplies the H_m recurrence, base rows, and sign polynomials, while `P15_S9C_KM_RECURRENCE_CERTIFICATE` supplies the corresponding K_m data. The inequalities above are exactly the cleared-denominator checks recorded by those certificates. $\square$

9 Proof Assembly

Proof of Theorem 1. The nonempty rows are exactly those of Lemma 3. For even n , Lemma 1 gives the required collapse, positivity, and ranks.

For odd $n = 2m + 1$, Lemma 2 reduces all nonempty rows with $k \geq 2$ to the even positive inputs and center trace identities. In the reference channel, the remaining $k = 1$ scalar is $a(2m, 1) + H_m$. Lemma 4 identifies H_m with the true center-nonfixed residual, and Lemma 6 gives $H_m > 0$.

In the pair-standard channel, the residual $k = 1$ eigenvalue is λ' , equivalently $K_m = (m-1)\lambda'$. Lemma 5 identifies this formula sequence with the true pair-standard scalar, and Lemma 6 gives $K_m > 0$. Thus both ambient standard-channel operators are nonzero scalar multiples of their classical matching-scheme idempotent components. Their ranks are therefore $\lceil n/2 \rceil$ and $\lceil n/2 \rceil - 1$. $\square$

10 Boundaries and Outlook

The theorem proved here is the scoped two-standard-channel rank theorem for the diagonal signed-reversal layer. The separator theorem is separately scoped to $X_n = \tilde{B}I_n(1, 1)$. The enumeration of positive fixed points and the matching-scheme multiplicities are used as classical background and are not claimed as new.

No Type-D theorem, full native B_n fingerprint theorem, all-irrep saturation theorem, or general Weyl-group theorem is asserted. The finite $n = 8$ fingerprint material in Appendix B is included because it clarifies the boundary of plausible stronger statements: it records exact finite zero mechanisms and selected finite defect accounting, while leaving the remaining atlas rows open.

The natural next mathematical stage is not to broaden the main theorem by wording, but to decide which finite fingerprint mechanisms admit clean all- n statements. Until such mechanisms are proved uniformly, they remain appendix evidence rather than theorem statements.

A Computational Certificates and Reproducibility

All paths in this appendix are relative to the companion repository root. The scripts are deterministic exact-arithmetic checks: they use integer or rational arithmetic and have no floating-point acceptance criterion. Artifact-producing scripts are rerun as

```
python -B <script> -write-json <json>.
```

When a script exposes Markdown output, the replay command also supplies `-write-md <md>`. Report-only commands omit output flags when the script does not expose them. The finite rows in these reports are edge checks or diagnostics; the all- m claims are carried by polynomial identities, integration-by-parts certificates, product/ Φ block accounting, and ratio induction.

Replay	Command script	Artifact basename	Status/role
S3	p15_s3_exact_engine.py	P15_S3_ENGINE_CERTIFICATE	PASS; engine
S4	p15_s4_even_theorem_certificate.py	P15_S4_EVEN_N_THEOREM_CERTIFICATE	PASS; even
S5	p15_s5_odd_structure_certificate.py	P15_S5_ODD_STRUCTURE_CERTIFICATE	PASS; odd split
S6	p15_s6_square_certificate.py	P15_S6_SQUARE_CERTIFICATE	PASS; counts only
S7	p15_s7_independent_red_team.py	P15_S7_INDEPENDENT_RED_TEAM_CERTIFICATE	PASS; replay
S8	p15_s8_hm_recurrence_certificate.py	P15_S8_HM_RECURRENCE_CERTIFICATE	PASS; H_m
S9	p15_s9_odd_pair_reduction_certificate.py	P15_S9_ODD_PAIR_REDUCTION_CERTIFICATE	PASS; pair red.
S9C	p15_s9c_km_recurrence_certificate.py	P15_S9C_KM_RECURRENCE_CERTIFICATE	PASS; K_m

Replay	Command script	Artifact basename	Status/role
S10	p15_s10_public_boundary_review.py	P15_S10_PUBLIC_BOUNDARY_CERTIFICATE	PASS; boundary
S11D	p15_s11_raw_rank_smoke_check.py	P15_S11_RAW_RANK_SMOKE_CHECK_CERTIFICATE	PASS; raw rank
S11A-H	p15_s11a_h_true_scalar_edge_insertion_verifier.py	P15_S11A_H_TRUE_SCALAR_EDGE_INSERTION_VERIFIER	PASS; H link
S11A-transfer	p15_s11a_scalar_link_symbolic_verifier.py	P15_S11A_SCALAR_LINK_SYMBOLIC_VERIFIER	PASS; transfer
S11A-K local	p15_s11a_k_true_scalar_bivariate_verifier.py	P15_S11A_K_TRUE_SCALAR_BIVARIATE_VERIFIER	PASS; local K
S11A-K global	p15_s11a_k_global_product_phi_certificate.py	P15_S11A_K_GLOBAL_PRODUCT_PHI_CERTIFICATE	PASS; global K
S12A	p15_s12_gate_a_signed_poset_certificate.py	P15_S12_GATE_A_SIGNED_POSET_CERTIFICATE	PASS; poset sep.
S12B	p15_s12_gate_b_tiler_certificate.py	P15_S12_GATE_B_TILER_CERTIFICATE	PASS; tiler obs.

For readability, the internal replay IDs have the following public roles: S11A-H is the odd reference true-scalar link; S11A-K local/global and S11A-transfer are the odd pair-standard true-scalar link; S8 and S9C are the holonomic recurrence/positivity certificates for the two residual sequences; S12A is the signed-poset separator certificate; and S12B is the translate-tiler divisibility certificate. The filenames are stable repository identifiers, not additional mathematical claims.

The rank theorem uses only the scoped two-standard-channel conclusion. The S12 rows add only the separately stated positive-only separator theorem for $X_n = \tilde{B}I_n(1,1)$, $n \geq 4$. The certificate table does not add a Type-D theorem, a full B_n fingerprint theorem, a classification theorem, a general Weyl theorem, or an enumeration novelty claim.

B Finite Native B_n Fingerprint Boundary

The main theorem of this paper is not a full native B_n Fourier fingerprint theorem. We nevertheless record one finite boundary computation because it is useful for interpretation: the naive expectation that every native irreducible block saturates the $R = +$ bound already fails at $n = 8$.

All statements in this appendix section are finite statements for the displayed $n = 8$ atlas. They do not imply an all- n classification, an all-irrep saturation theorem, or a general Johnson-scheme vanishing theorem. The external $n = 8, k \geq 2$ defect census contains 41 fixed-plus family-instances and 114 defect rows. The currently certified accounting covers 6 family-instances and 44 rows, leaving 35 family-instances and 70 rows open.

Slice	fam.	rows	acct. fam.	acct. rows	open fam.	open rows
$n = 8, k = 2$	5	20	1	15	4	5
$n = 8, k = 3$	36	94	5	29	31	65
total	41	114	6	44	35	70

The accounted family-instances are

$$(n, k, \lambda^+) \in \{(8, 2, (1)), (8, 3, (1, 1)), (8, 3, (2)), (8, 3, (4)), (8, 3, (4, 1)), (8, 3, (2, 2, 2, 1, 1))\}.$$

The phrase “accounted” here means finite exact certificate or finite quotient-layer accounting, not an all- n theorem.

Proposition 1 (Pre-Specht cancellation for the $(1,1)$ zero family). *For $X = \tilde{B}I_8(3,3)$ and every partition $\nu \vdash 6$,*

$$M_{(1,1)|\nu}(X) = 0.$$

Certified proof sketch. Before applying a complement Specht module, the fixed-plus block with $\lambda^+ = (1, 1)$ is a matrix over the group algebra $\mathbb{Q}[S_6]$. The exact certificate groups the 192 active signed terms into 96 group-algebra keys. For every key there are two surviving completions with the same complement permutation and opposite S_2 sign; their coefficients are -4 and 4 . Thus all 96 keys vanish already in $\mathbb{Q}[S_6]$. Applying any representation S^ν , $\nu \vdash 6$, preserves zero. $\square$

Proposition 2 (Pure-plus Johnson degree-drop). *For $X = \tilde{B}I_8(3, 3)$,*

$$M_{(2,2,2,1,1)|\emptyset}(X) = 0.$$

Certified proof sketch. The conjugate partition of $(2, 2, 2, 1, 1)$ is $(5, 3)$, so

$$S^{(2,2,2,1,1)} \cong S^{(5,3)} \otimes \text{sgn}.$$

After the corresponding sign twist, the certified operator A_{sgn} acts on the 3-subset permutation module $M^{(3)}$. Let U_2 be the pair-incidence submodule spanned by the vectors v_P of 3-subsets containing a fixed pair P . The Johnson certificate verifies the exact containment

$$A_{\text{sgn}}(M^{(3)}) \subseteq U_2.$$

Since the top quotient $M^{(3)}/U_2$ is isomorphic to $S^{(5,3)}$, the sign-twisted operator kills this quotient. Untwisting by the sign representation gives the displayed pure-plus zero. An independent direct rational seminormal Specht computation also gives zero entries throughout the 28-dimensional block. $\square$

Remark 2 (Appendix boundary). *The finite zero mechanisms above explain all twelve zero rows under the $n = 8, k = 3$ atlas census: eleven rows from Proposition 1 and one row from Proposition 2. Other certified finite packages account for selected nonzero extra-defect families, including $T_1^{(8,2)}$, $T_2^{(8,3)}$, $T_4^{(8,3)}$, and $\lambda^+ = (4, 1)$ at $n = 8, k = 3$. The remaining extra-defect rows are not classified here.*

Data and Code Availability

The companion repository contains the LaTeX source, deterministic replay scripts, certified JSON/Markdown artifacts, finite atlas inputs used by the appendix, and a SHA-256 manifest for the public package. The scripts replay the finite checks and exact polynomial identities referenced in Appendix A; the theorem statements are the statements proved in this manuscript.

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